

생성형 AI란

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생성형 AI (Generative AI)

➤ 생성형 AI란?

- 생성형 AI(Generative AI)는 AI를 활용하여 사용자의 입력을 기반으로 텍스트, 이미지, 음악 등의 콘텐츠를 생성할 수 있는 AI 모델

➤ 기존 AI와 생성형 AI의 차이는?

- 사용되는 기술은 동일
- 기존 AI는 많은 양의 데이터를 활용하여 '특징'을 학습하고 이를 기반으로 '예측'을 수행
- 생성형 AI는 데이터를 학습 후 '새로운 창작물'을 만듦



생성형 AI (Generative AI)

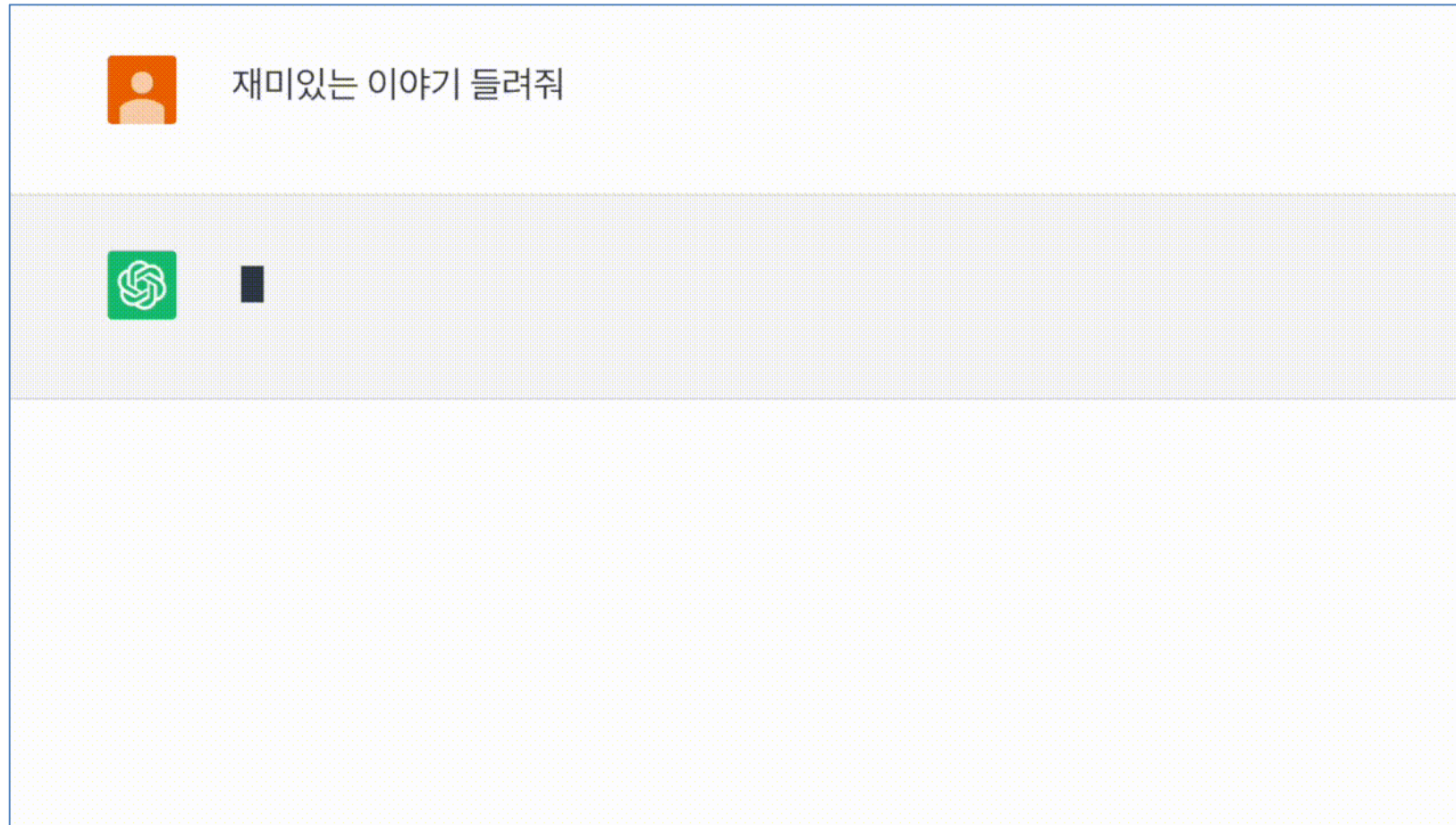
➤ AI가 만든 이미지, 동영상

- 스페이스 오페라 극장: 2022년 8월 미국 콜로라도 주립 박람회에서 열린 한 미술대회에 신인 디지털 아티스트 부문 1위를 차지



생성형 AI (Generative AI)

➤ AI가 들려 주는 이야기



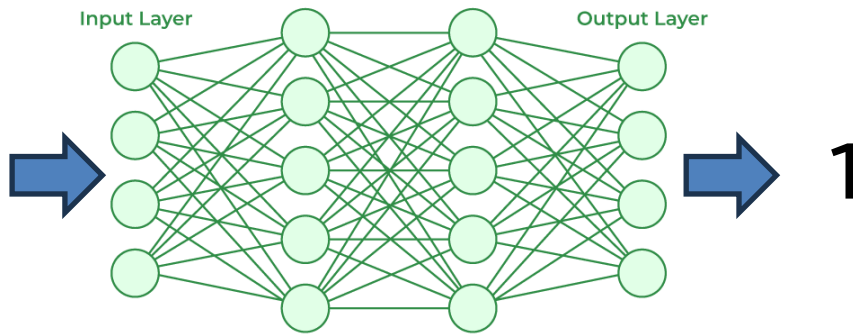
생성형 AI (Generative AI)

➤ AI가 만든 이미지, 동영상

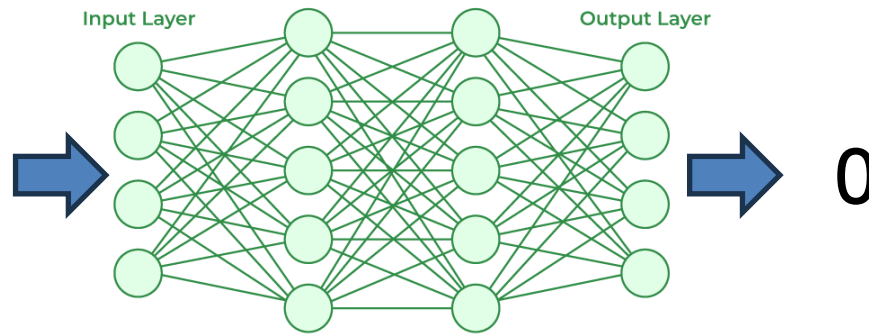


Predictive vs Generative

➤ Predictive Model



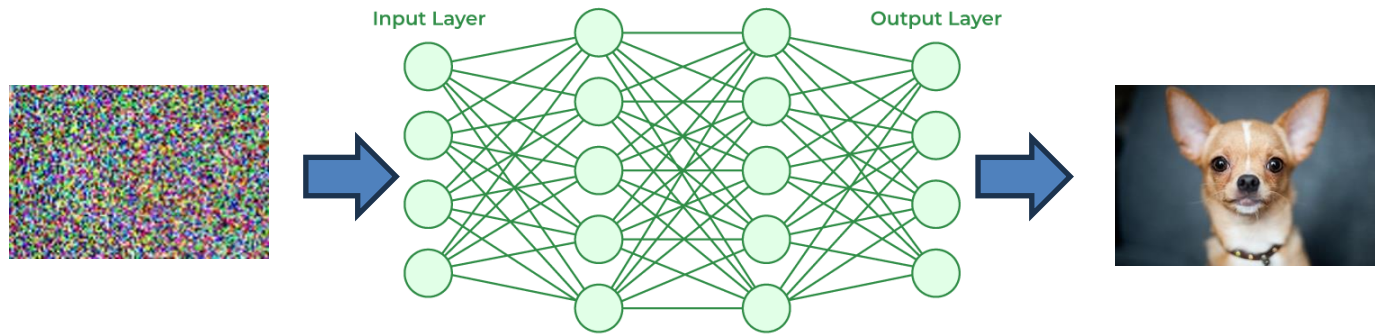
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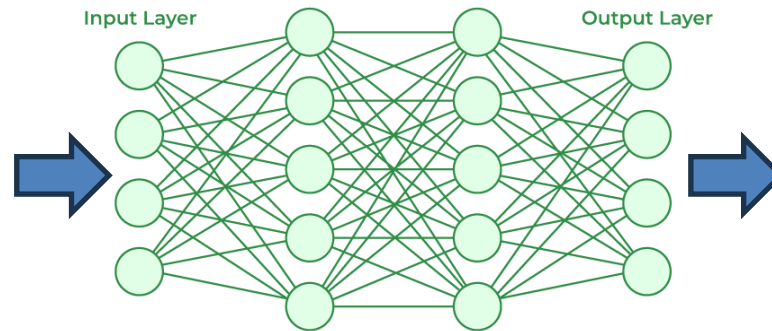
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Predictive vs Generative

➤ Generative Model



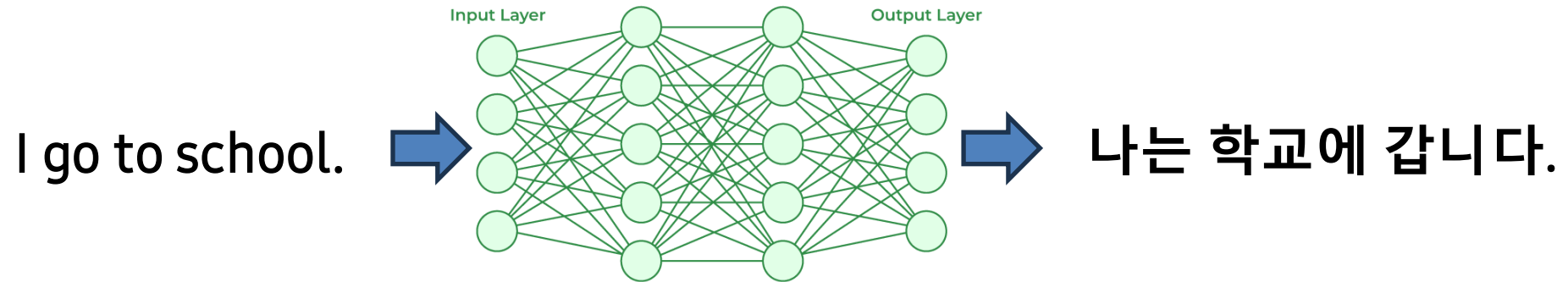
I go to school



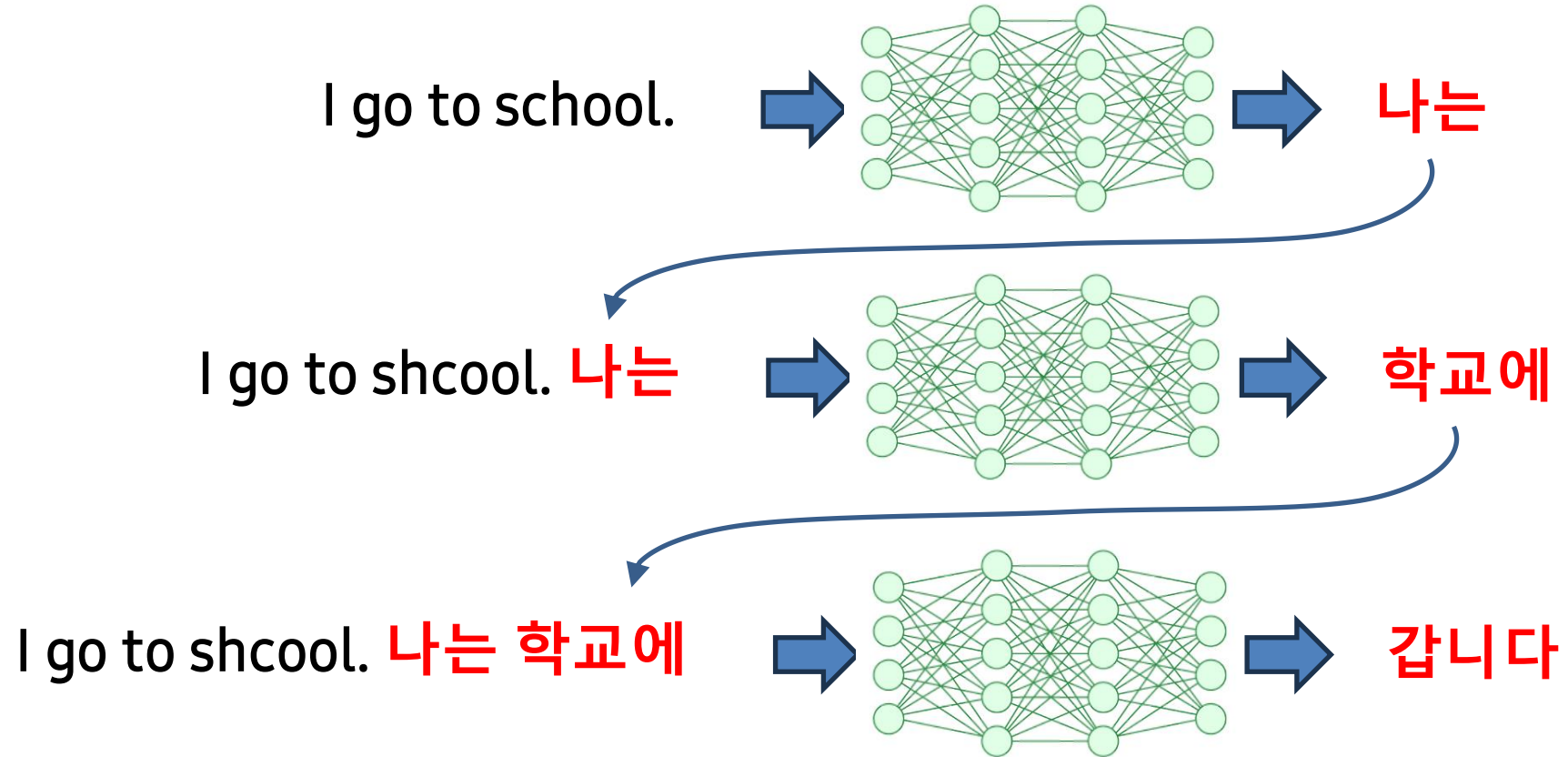
나는 학교에 갑니다.

2. 언어 생성형 모델

➤ 언어 생성형 모델



➤ 언어 생성형 모델



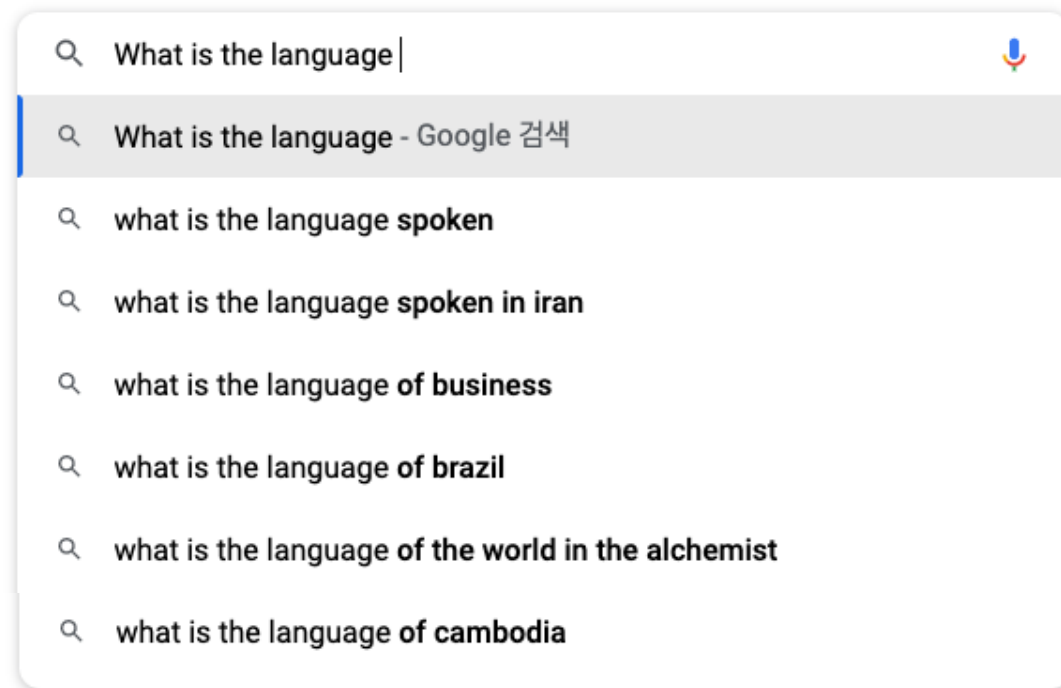


Language Models predict the next word for a given sentence

I go to school by ?

I go to church on ?

➤ You use Language Models everyday!

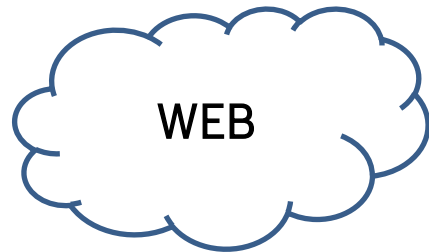


언어 모델 만들기

➤ 5개 단어를 보고 다음 단어 예측하기: 어떻게 만들면 될까?

- 웹에서 “I go to school by”로 시작하는 문장 모으기

I go to school by ?



I go to school by bus.
I go to school by car.
I go to school by walk.
I go to school by train.
I go to school by bus.
I go to school by bus.
I go to school by walk.

등장 확률기반으로
하나의 단어를 선택

언어 모델 만들기

➤ 5개 단어를 보고 다음 단어 예측하기: 어떻게 만들면 될까?

- 이것을 테이블로 만들어서 저장

5단어 문장	다음 단어의 확률								
	bus	train	car	walk	on	in	Mon.	.	,
I go to school by	0.3	0.1	0.1	0.4	0	0	0	0	0
go to school by bus	0	0	0	0	0.3	0.2	0	0.4	0.1
to school by bus on	0	0	0	0	0	0	0.2	0	0
school by bus on Mon	0	0	0	0	0	0.2	0	0.2	0.1
...	...	...	...	...	...	...	...	...	...

➤ 몇가지의 문제

- 테이블의 크기는?
- 테이블에 없는 패턴이 나오면?

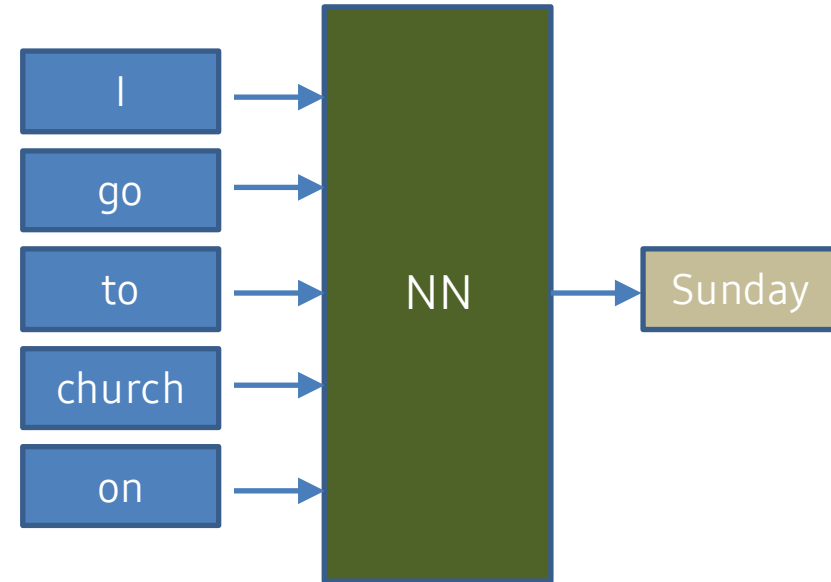
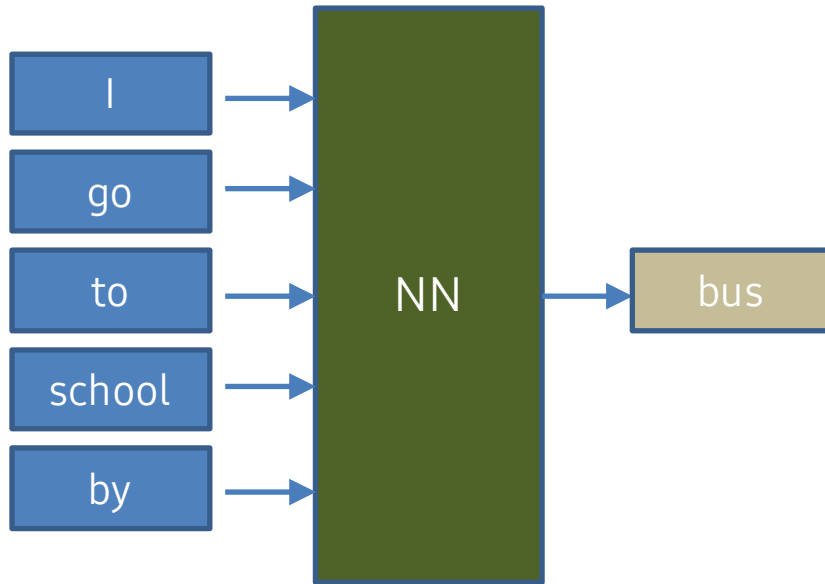


Neural Network으로 해결

- 엄청남 암기력
- 배운 것의 일반화

Neural Language Model

➤ Language Model with Neural Network



Neural Language Model

➤ Translation with Next Word Prediction

Parallel Dataset

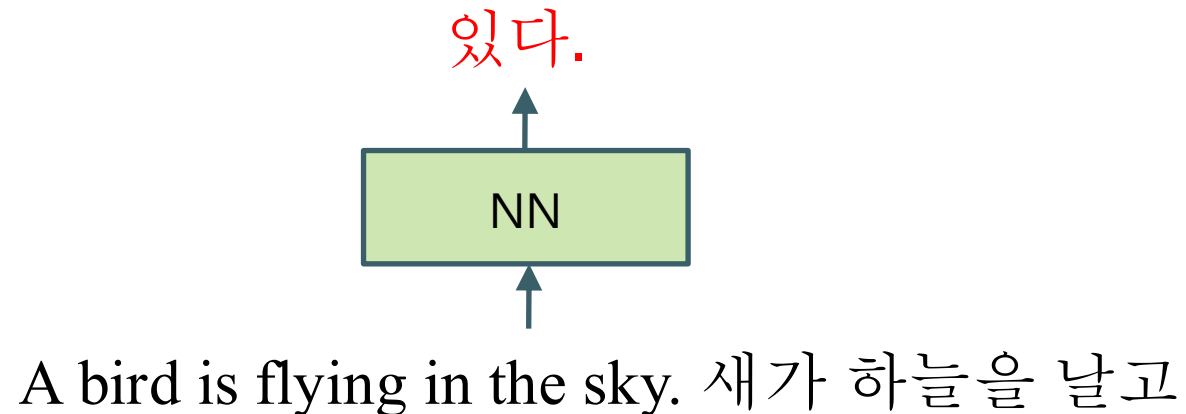
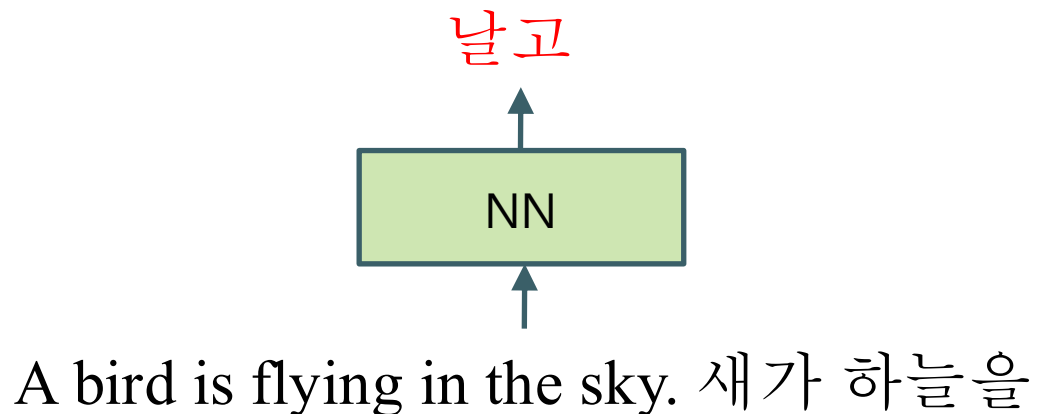
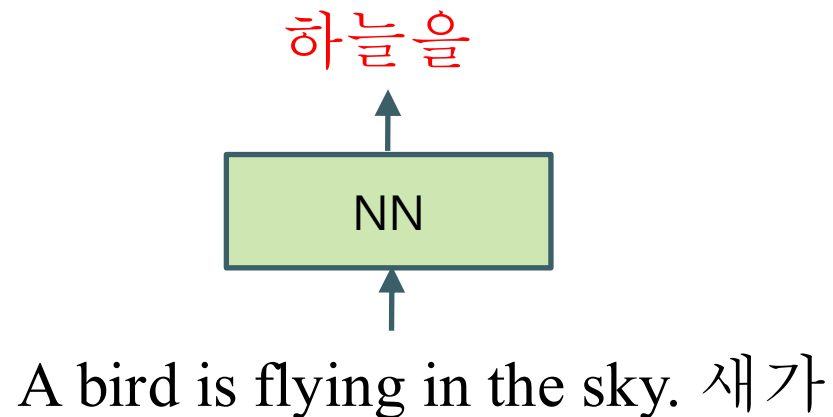
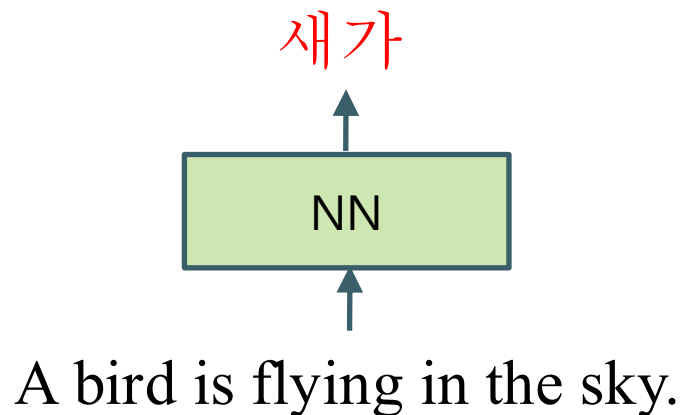
A bird is flying in the sky. 새가 하늘을 날고 있다.
A dog is running on the field. 개가 운동장에서 뛰고 있다.



(A bird is flying in the sky. → 새가)
(A bird is flying in the sky. 새가 → 하늘을)
(A bird is flying in the sky. 새가 하늘을 → 날고)
(A bird is flying in the sky. 새가 하늘을 날고 → 있다.)
(A dog is running on the field. → 개가)
(A dog is running on the field. 개가 → 운동장을)
...

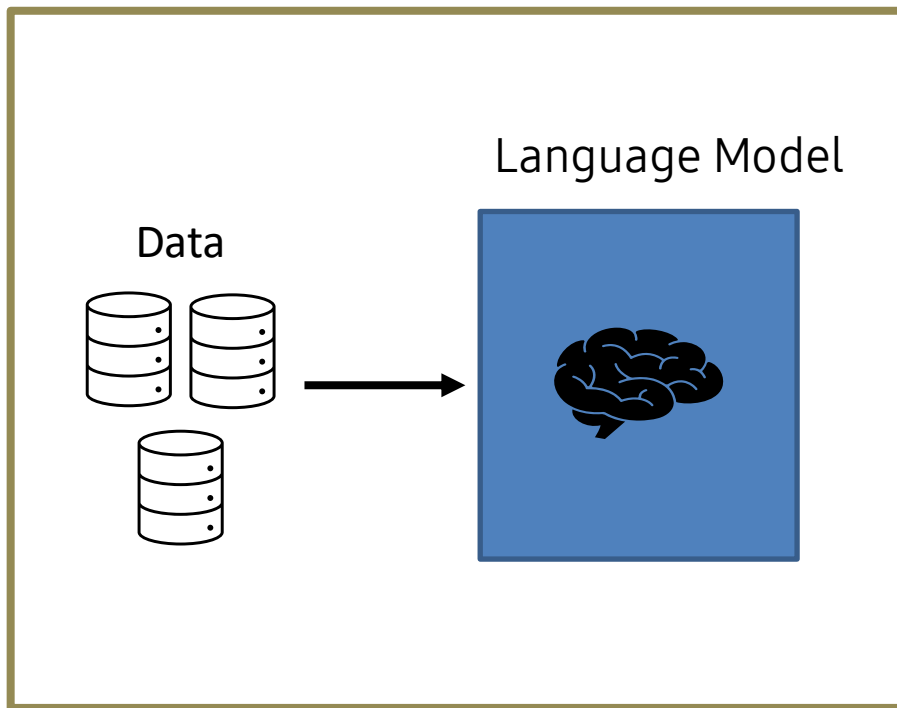
Neural Language Model

➤ Translation with Next Word Prediction



Neural Language Model

➤ In-Context Learning

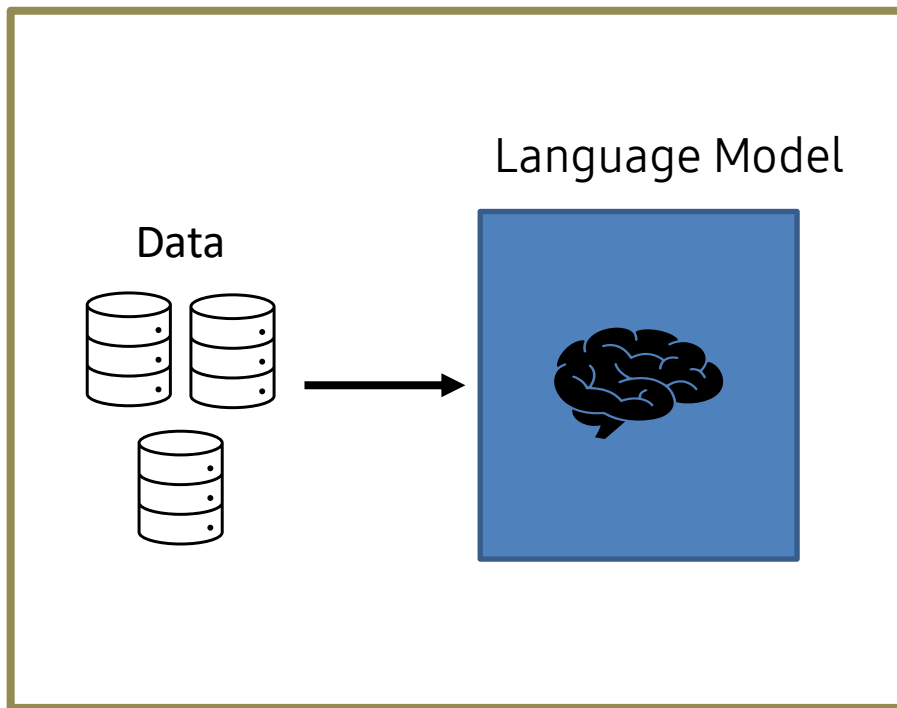


많은 일반 데이터를 모아서 생성된
Next word prediction 모델

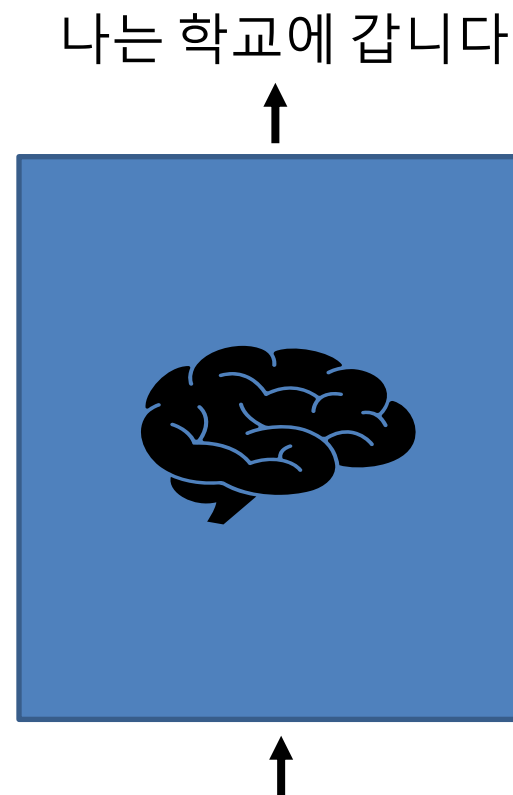


Neural Language Model

➤ In-Context Learning



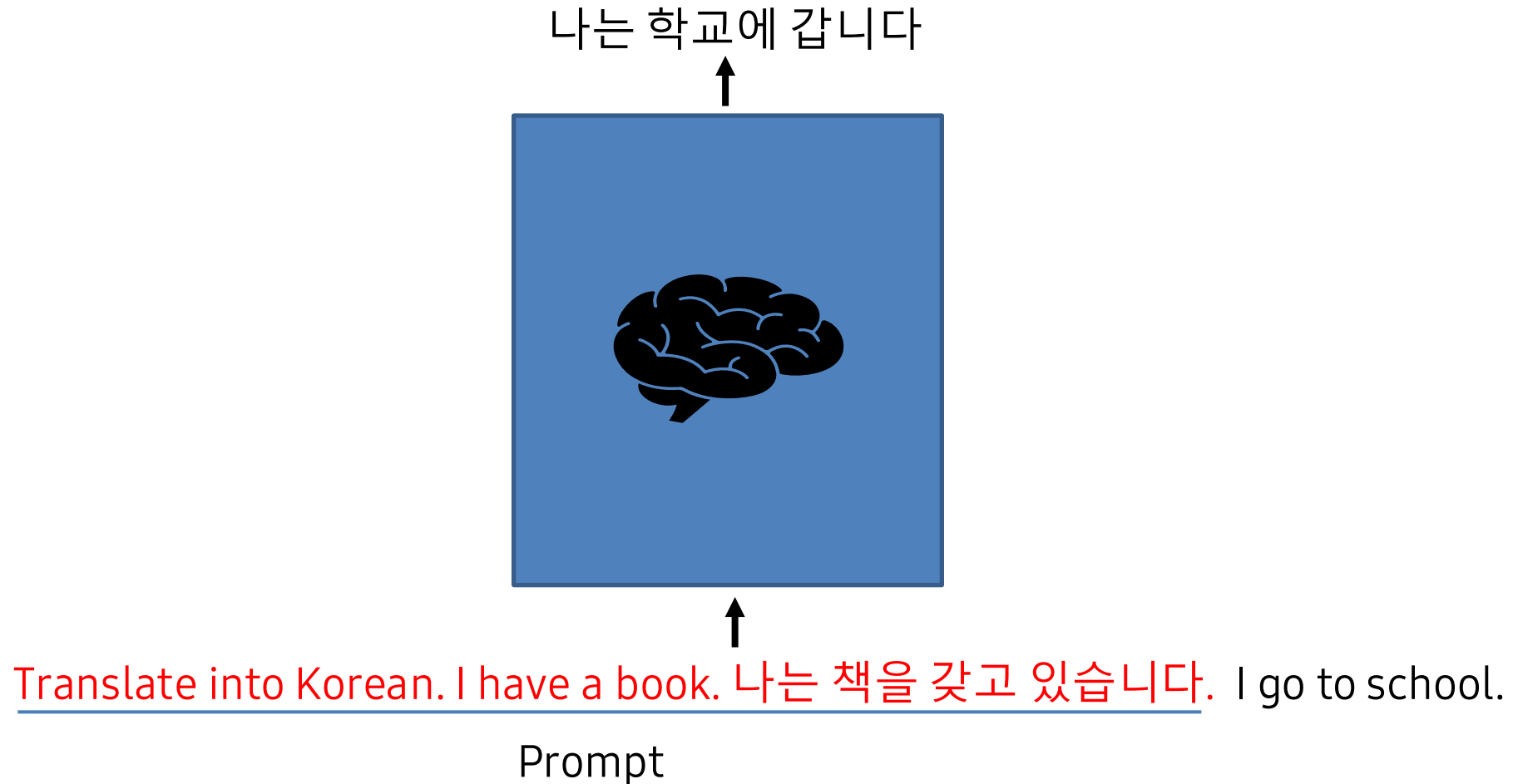
많은 일반 데이터를 모아서 생성된
Next word prediction 모델



Translate into Korean. I go to school

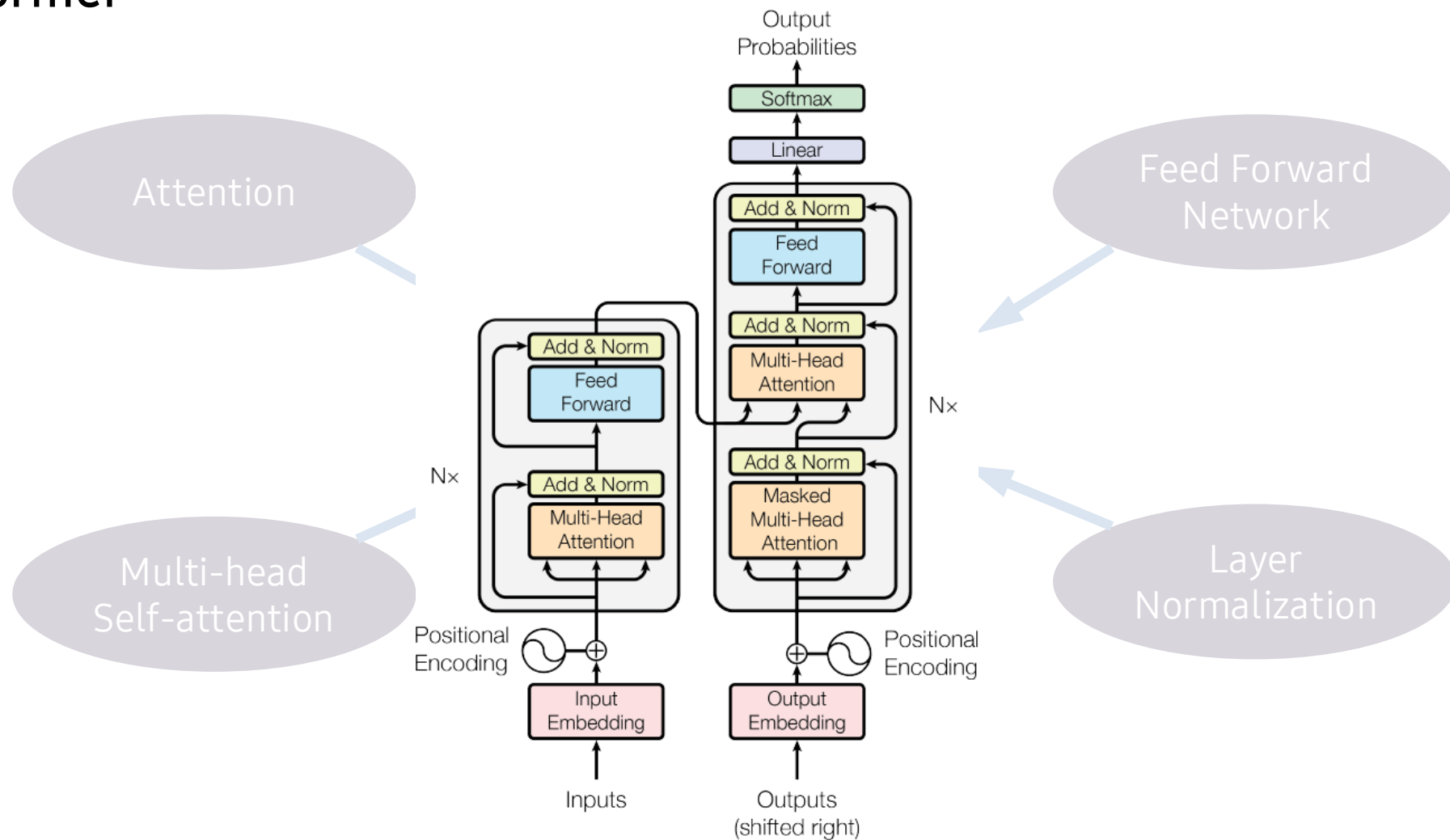
Neural Language Model

➤ In-Context Learning

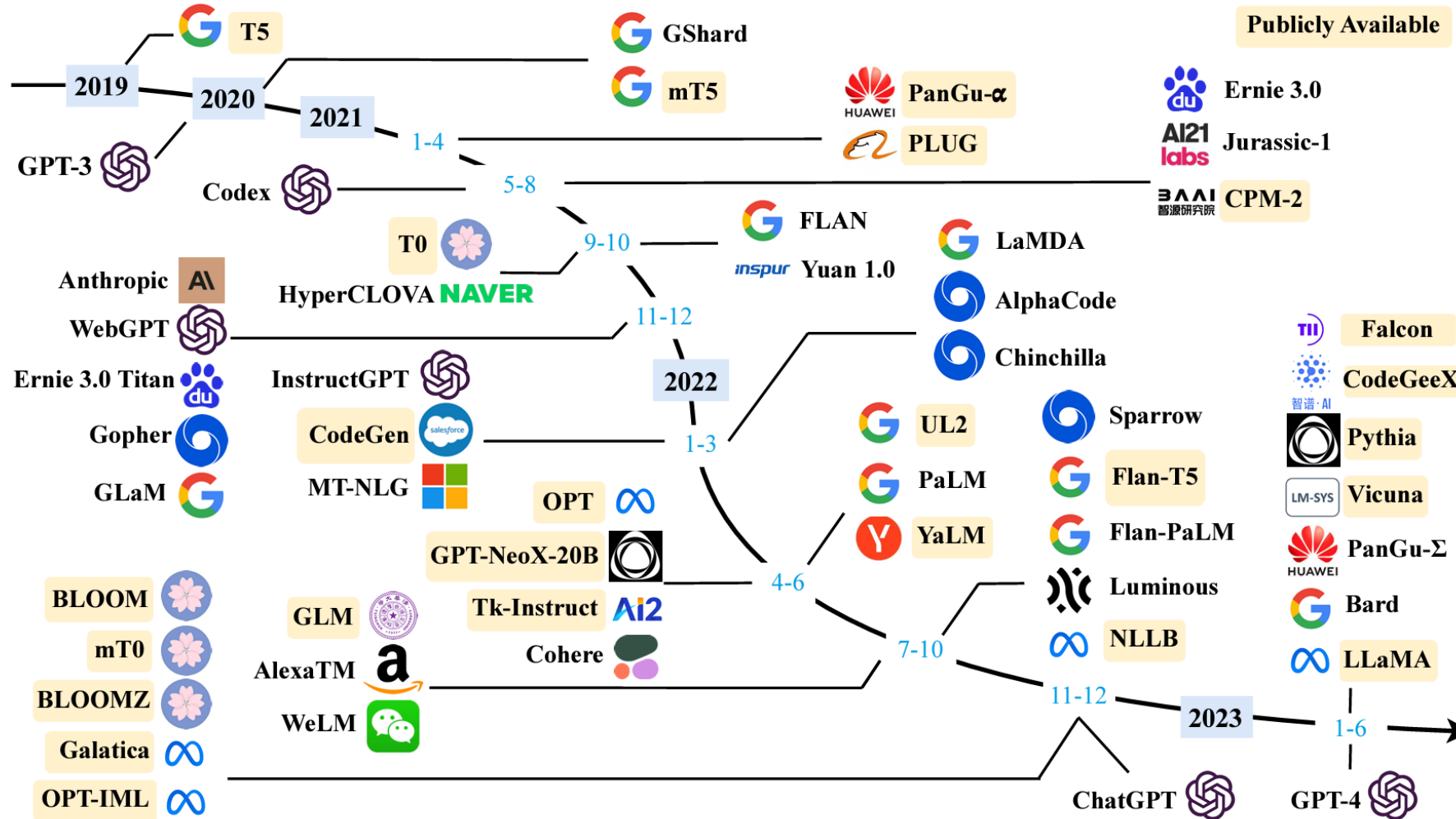


Neural Language Model 종류

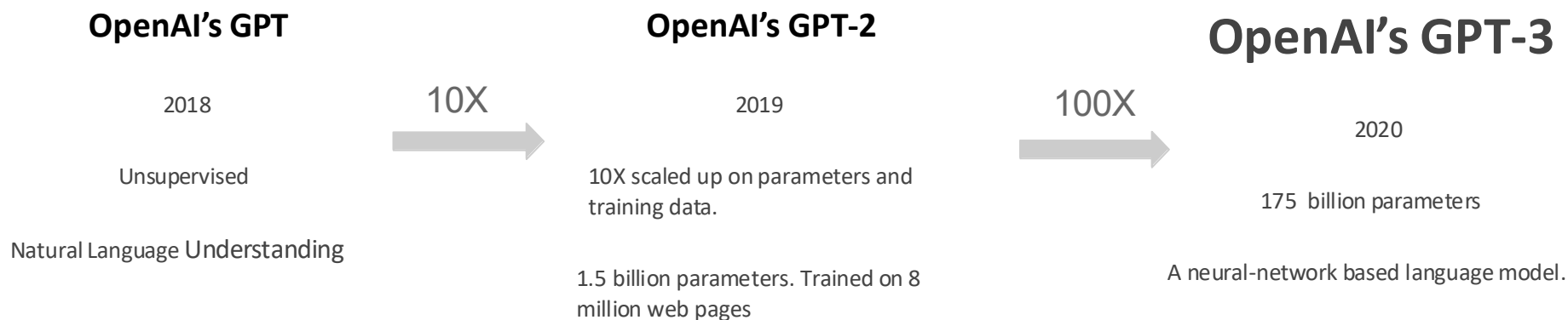
➤ Transformer



Neural Language Model 종류

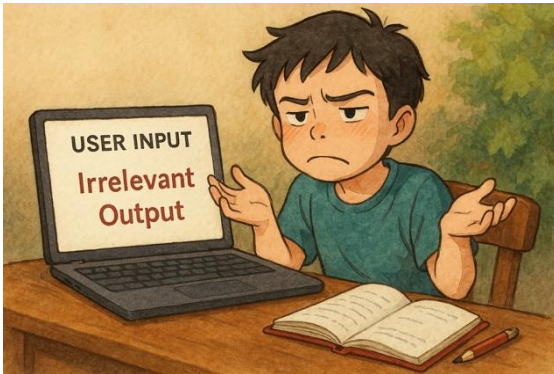


➤ Generative Pretrained Transformer (GPT) models



언어 모델의 문제점

- **유용성 부족:** 사용자의 입력에 부합하지 않는 답변 생성
- **환각:** 존재하지 않거나 잘못된 사실을 만듦
- **해석 가능성 부족:** 모델이 결과를 출력한 이유를 설명하기 어려움
- **편향 또는 독성 출력:** 남녀, 인종 등의 편향, 독성이 있는 답변 출력 가능성



왜 이런 문제가 생길까?

The output is not aligned with its users

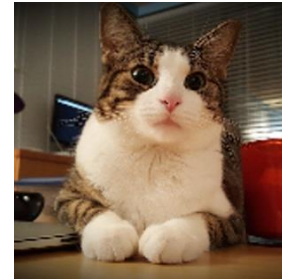
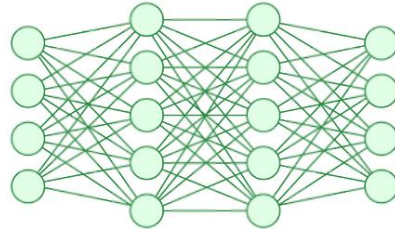
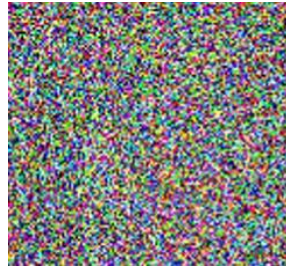
- 웹에서 수집된 많은 데이터를 사용했지만, 정제되지 않은 데이터
 - 수 많은 편향된 문장, 저속어 등이 포함될 가능성이 높음
- 최신 데이터는 학습에 포함되지 않으므로, 적절한 답을 생성하지 못함
- 단순히 확률 기반의 next word prediction이 인간이 원하는 답이 아닐 수 있음

- RLHF: Reinforcement Learning with Human Feedback
- RAG: Retrieval-Augmented Generation

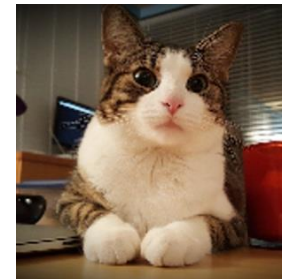
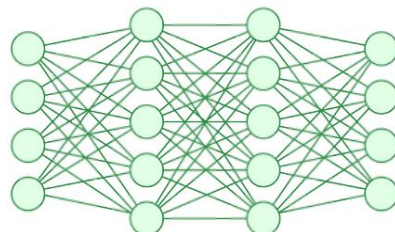
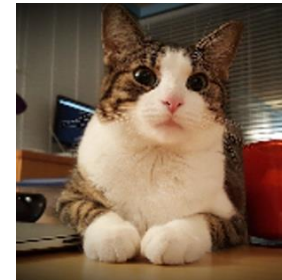
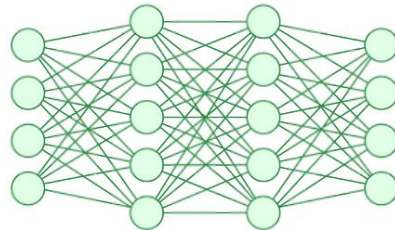
4. 이미지 생성형 모델

이미지 생성 모델이란

➤ Image Generative



A cat sitting
on the floor

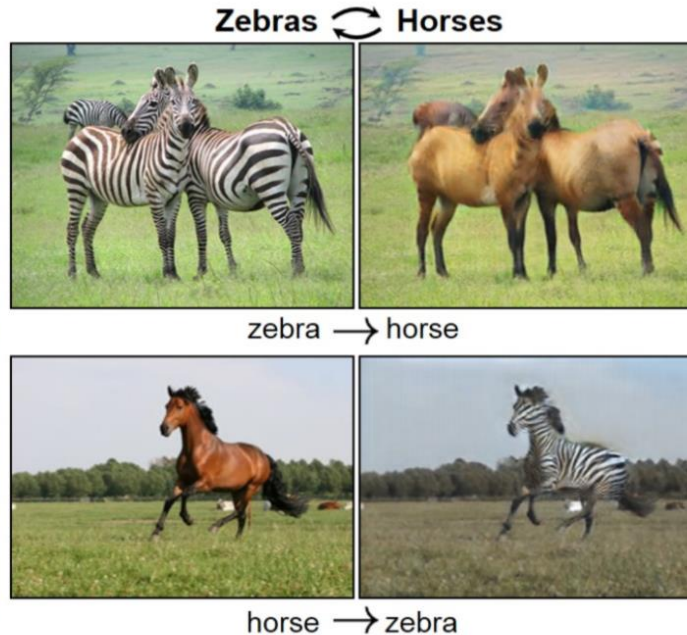


이미지 생성 모델이란

- Generate new samples from the same distribution with training data
 - Super-resolution, Style transfer, Image inpainting, Image generation



Super-resolution [Ledig, et. al., 2017]



Style transfer [Zhu, et. al., 2017]



High-res image generation
[Karras, et. al., 2018]

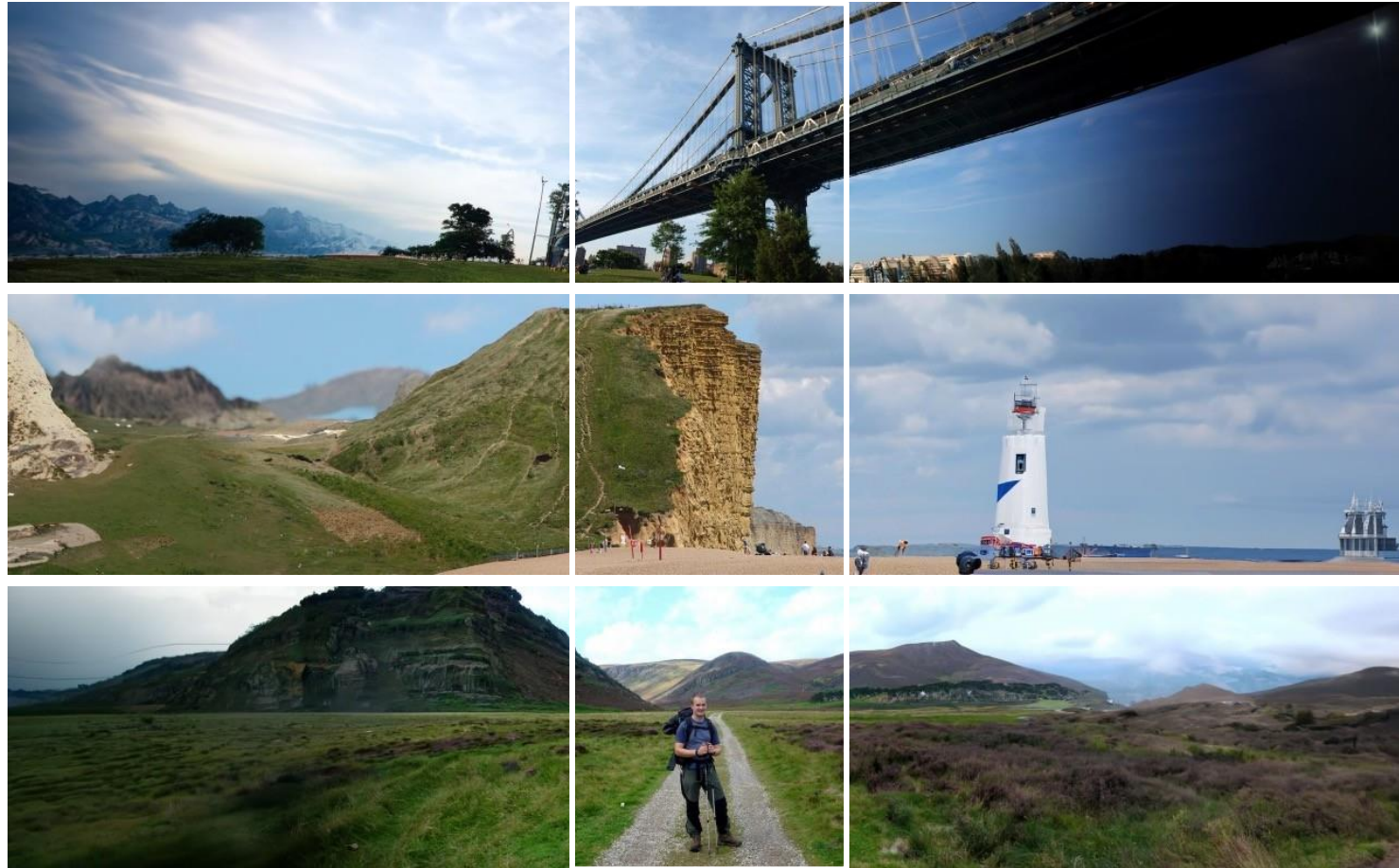
이미지 생성 모델이란

➤ AI Art



이미지 생성 모델이란

➤ Image Outfilling

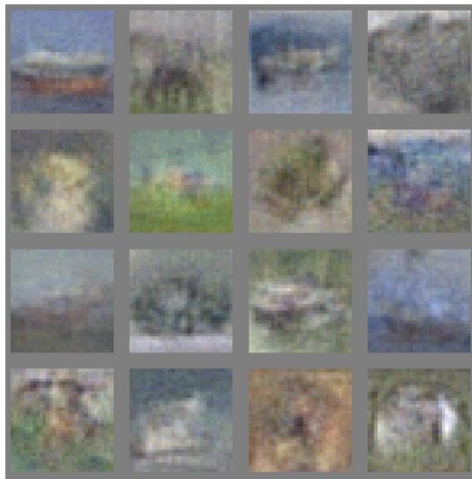


여러가지 이미지 생성 기법

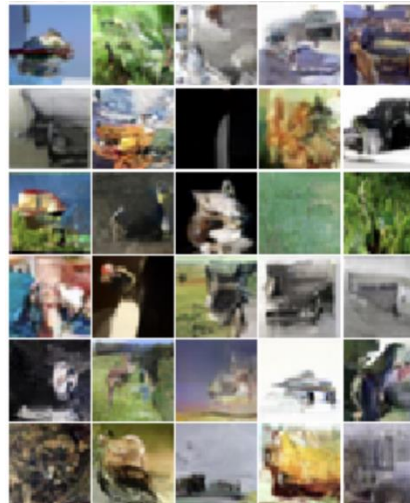
VAEs, 2013



GANs, 2014



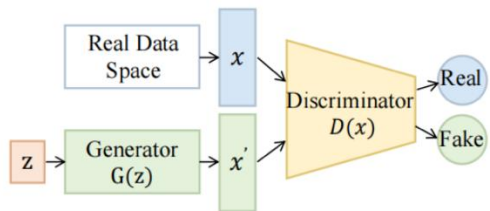
PixelCNN, 2016



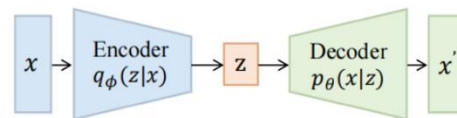
BigGAN, 2019



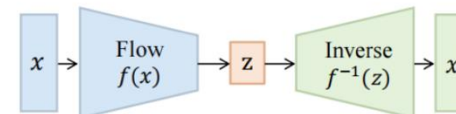
Imagen, 2022



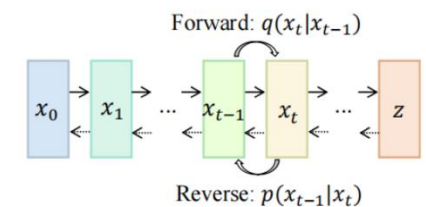
(1) Generative adversarial networks



(2) Variational autoencoders



(3) Normalizing flows



(4) Diffusion models

여러가지 이미지 생성 기법

Variational Autoencoders

Autoregressive Models

Normalizing Flows

Generative Adversarial Networks

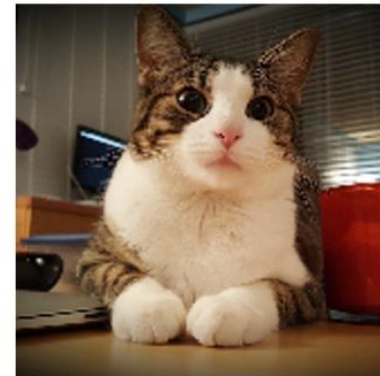
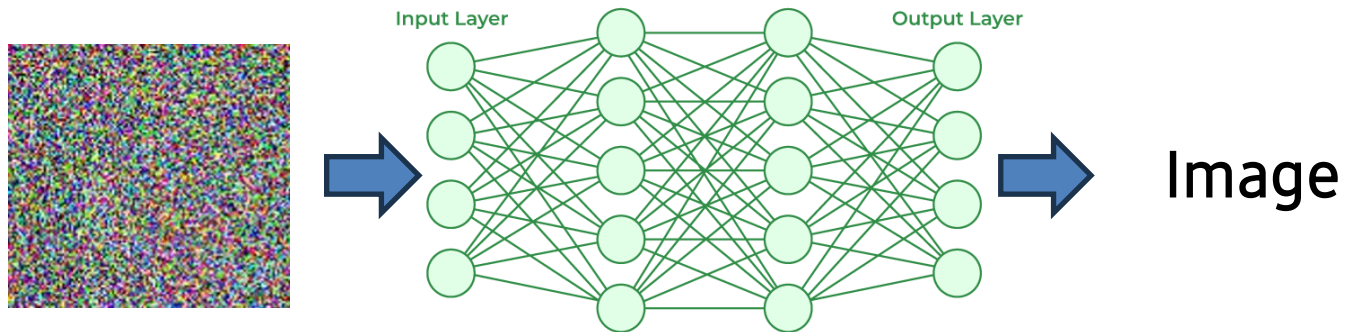
Energy-based Models

Diffusion Models



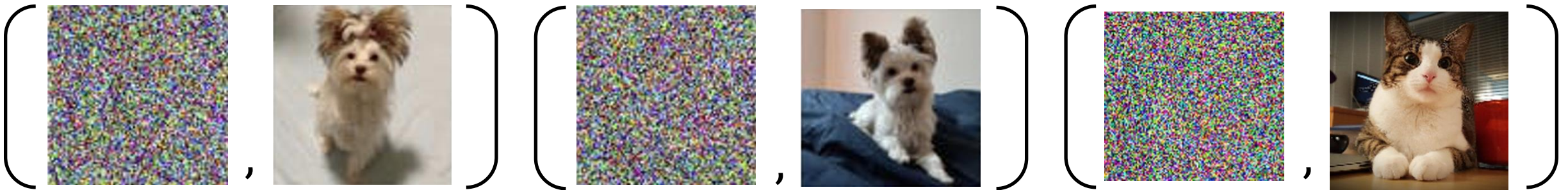
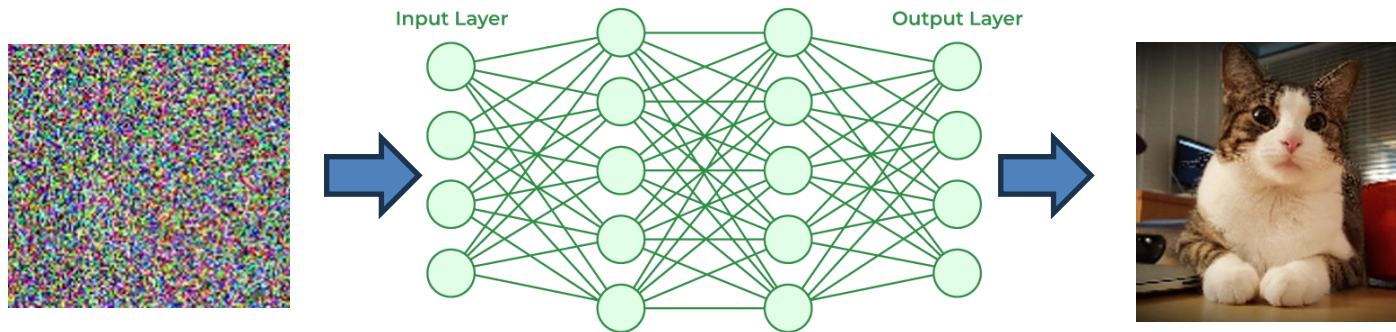
이미지 생성 AI 모델이란

➤ Naïve Image Generative Model



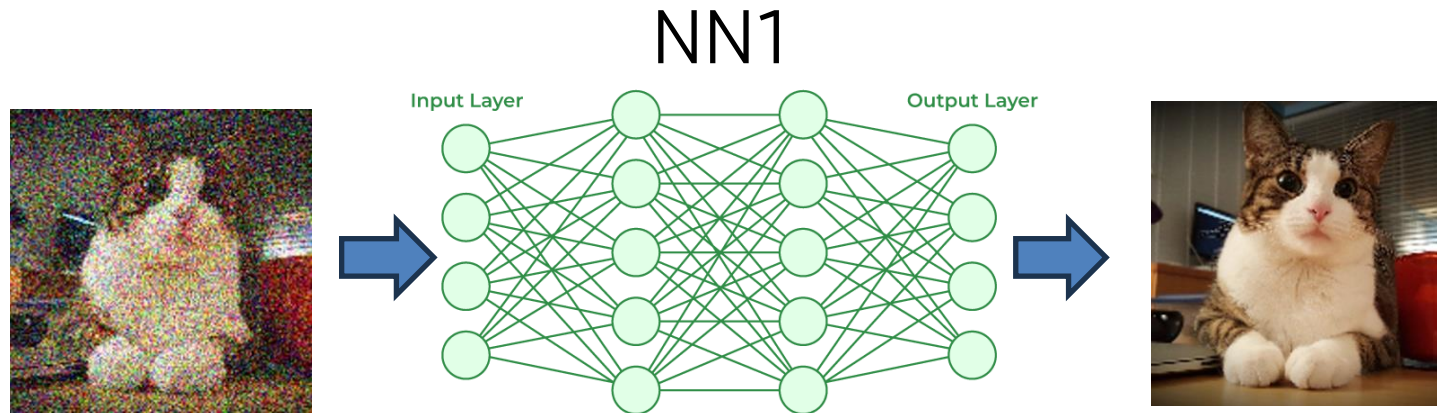
이미지 생성 AI 모델이란

➤ Naïve Generative Model



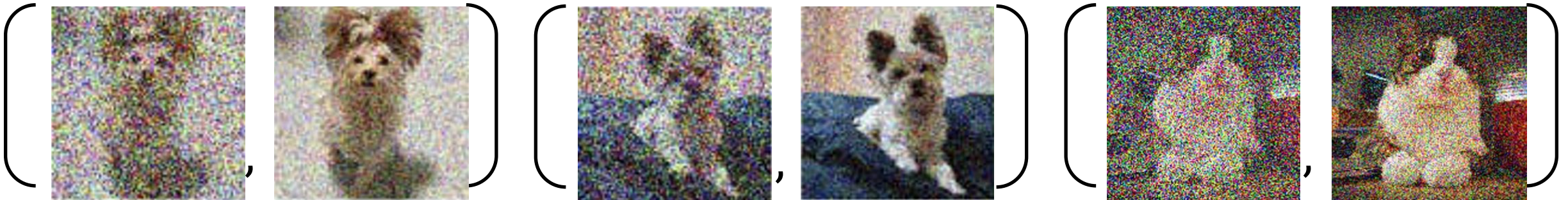
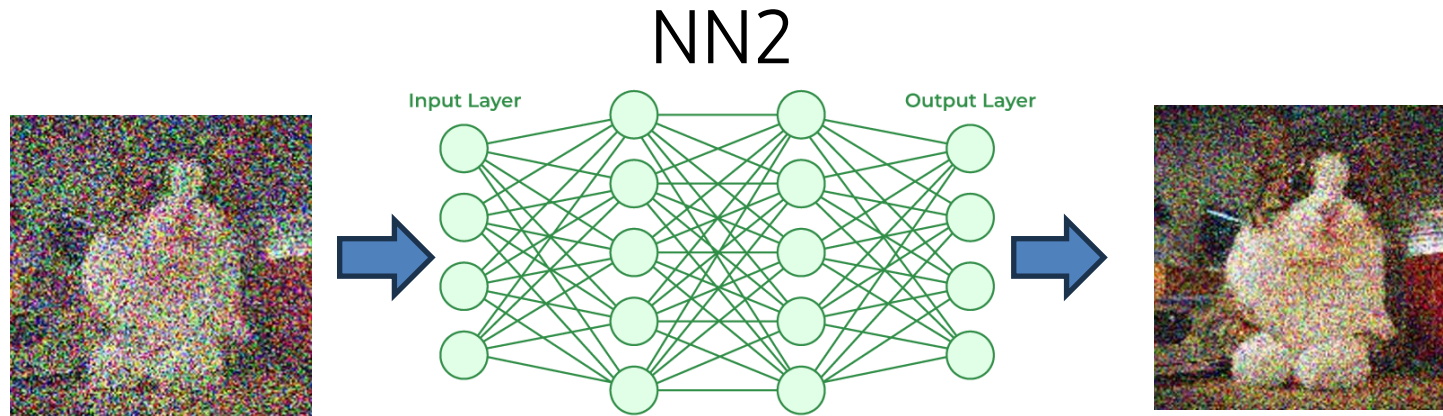
Diffusion Model

➤ De-noising Models



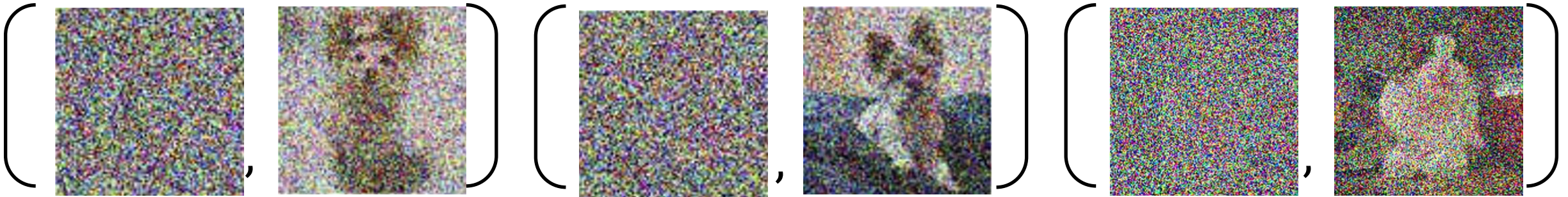
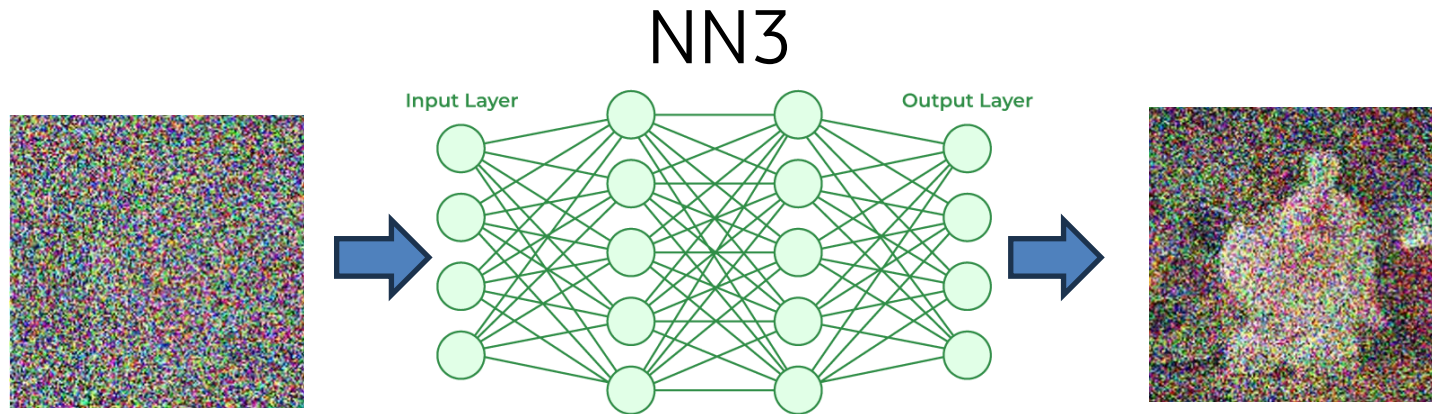
Diffusion Model

➤ De-noising Models



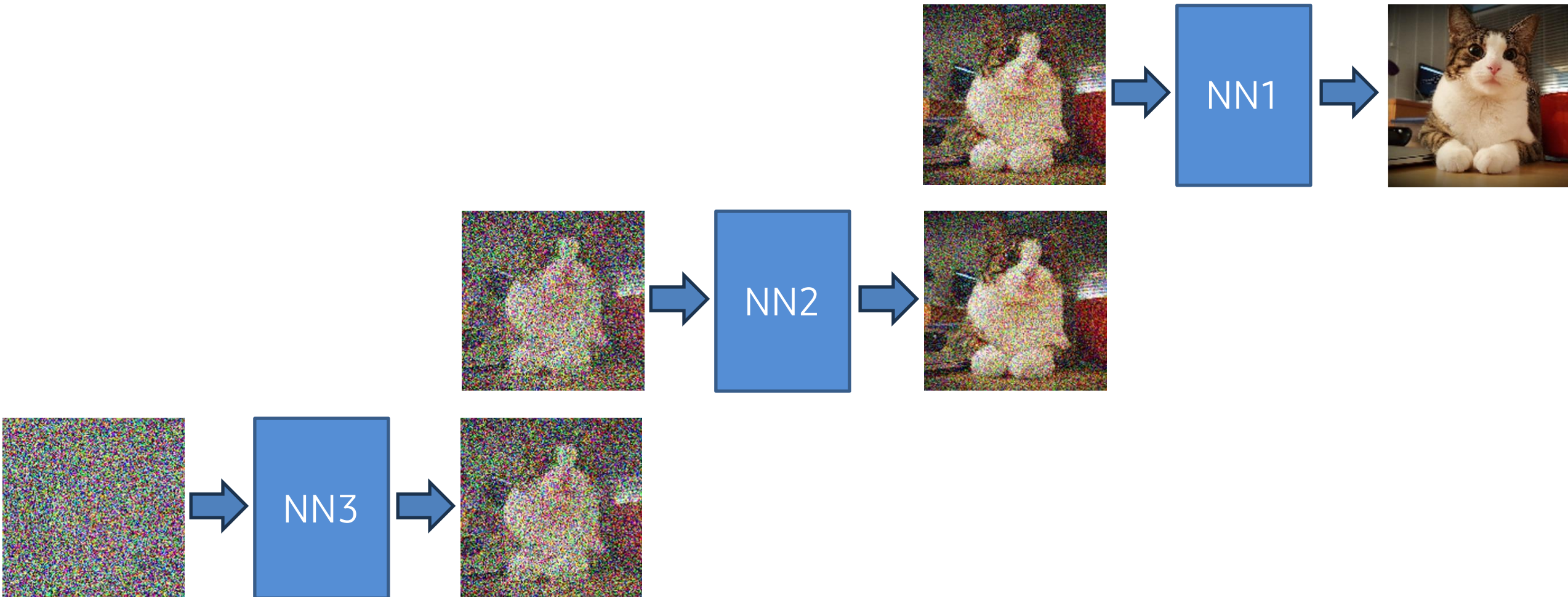
Diffusion Model

➤ De-noising Models



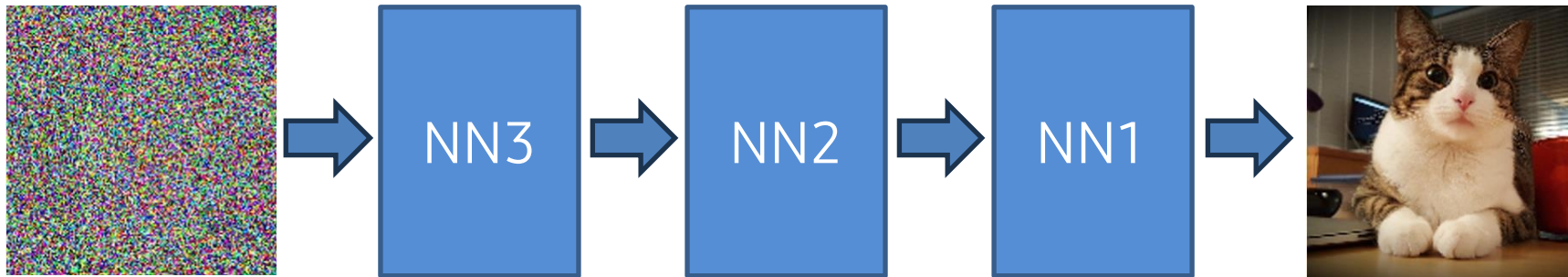
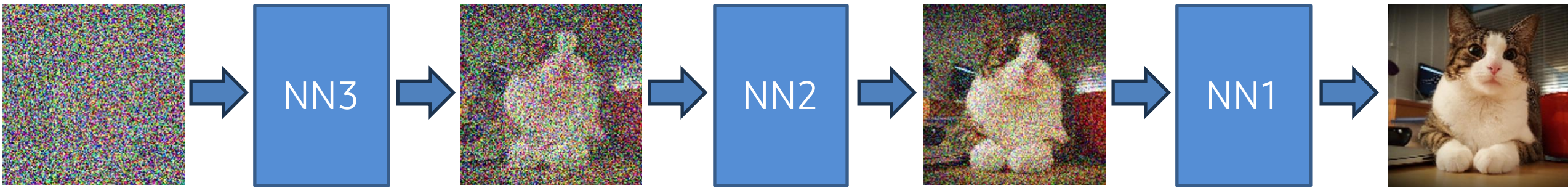
Diffusion Model

➤ Cascade of De-noising Models



Diffusion Model

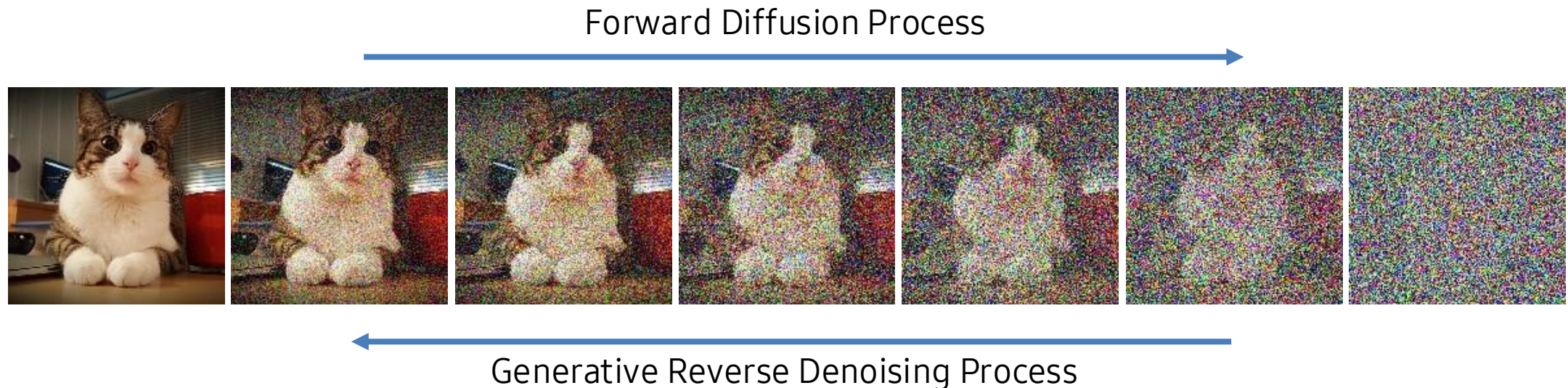
➤ Cascade of De-noising Models



Diffusion Model

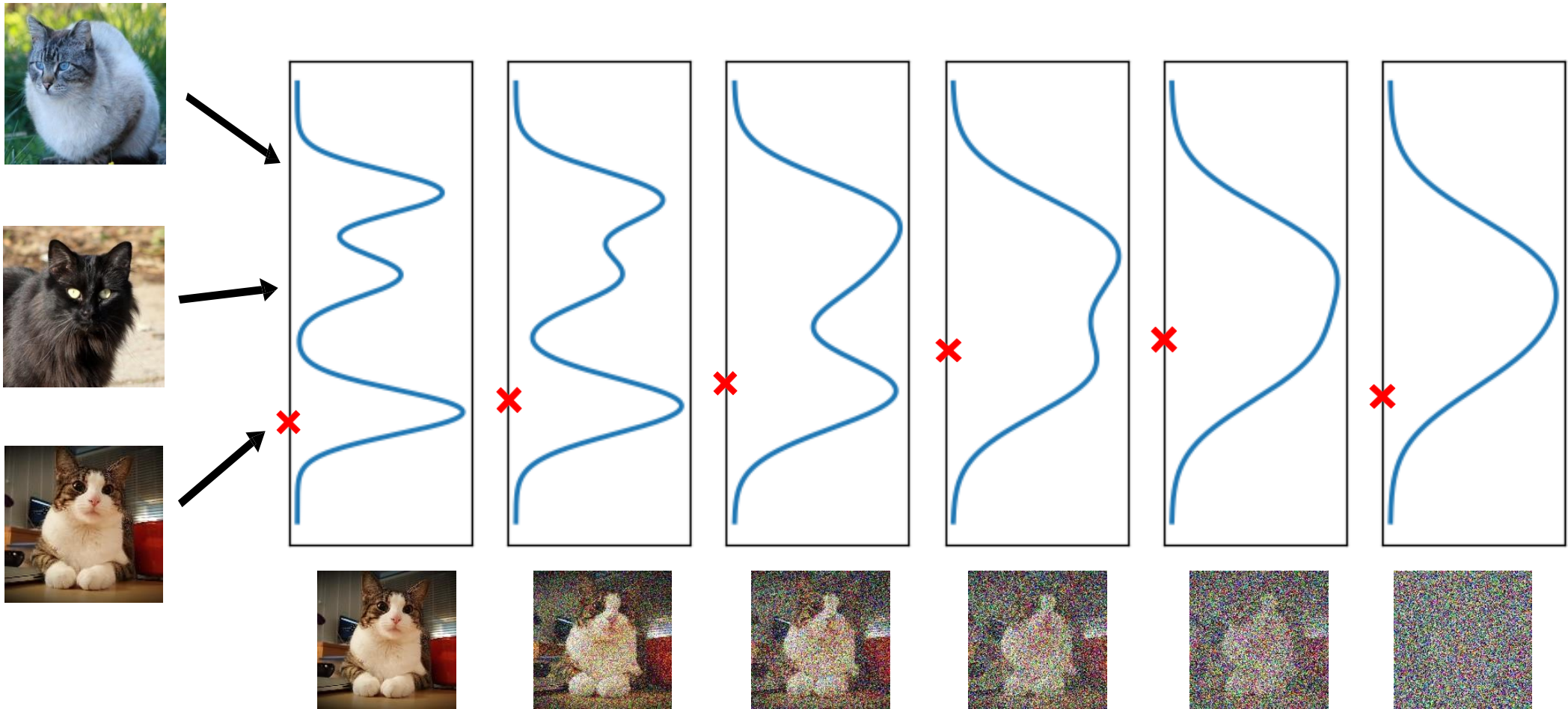
➤ 학습과정 요약

- 이미지에 노이즈를 추가하는 과정을 반복하여 진행
- 이를 반대로 노이즈로부터 이미지로 복원하는 방법을 모델이 학습
- 결과적으로 랜덤 노이즈로부터 이미지를 생성해 내는 과정을 모델이 학습함



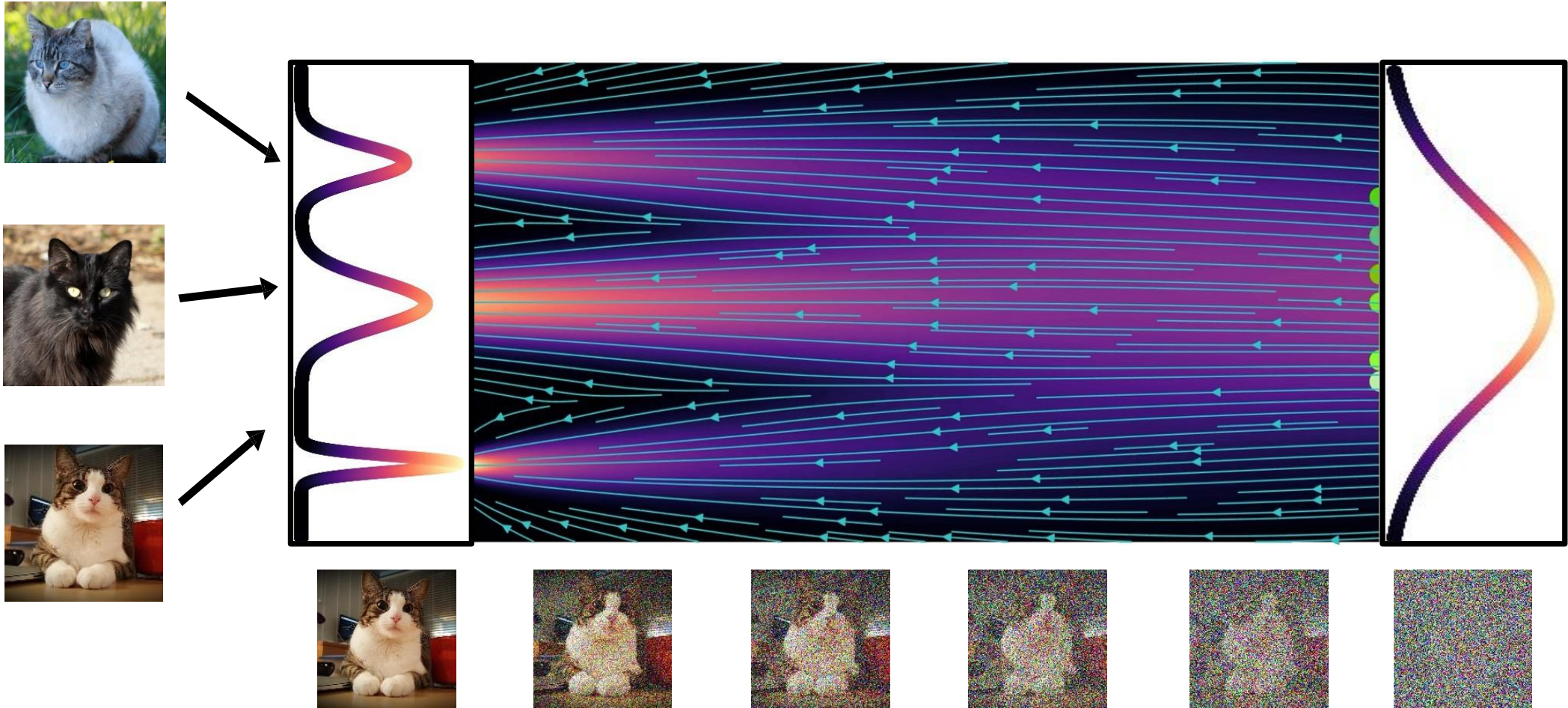
Diffusion Model

➤ 왜 Diffusion이라 불리는가?



Diffusion Model

➤ 왜 Diffusion이라 불리는가?



Diffusion Model

➤ 장점

- 기존의 생성모델보다 사실적이고 정교한 이미지를 생성함
- 학습 과정이 안정적이며 다양한 이미지 생성이 가능함

➤ 단점

- 학습 과정이 여러 스텝으로 구성됨에 따라 학습 속도 및 생성 속도가 느림
(최근개선 연구가 지속적으로 발표되며 해당 단점이 많이 해결됨)

